Apoorva BUSUNUR MALLIKARJUNA

MSc in Aerospace Engineering (Embedded Systems) — ISAE-SUPAERO, Toulouse, France

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Profile — Analytical engineer with experience in **embedded systems** and **EEG-based blink detection** for fatigue assessment ensuring **operator safety** in extreme environments. Interested in gaining research experience in the domains: **neuro-adaptive systems**, **brain-computer interfaces**, and **human factors** for aerospace applications. Seeking Research Engineer/Assistant opportunities from September 2026.

Experience

Research Intern, Neuroergonomics Lab, ISAE SUPAERO, Toulouse **Apr 2025 – Sep 2025**

- Developed real-time blink detection algorithms using fronto-polar EEG data to assess mental fatigue in professionals working in extreme environments.
- Implemented an experimental setup and conducted robustness analysis to ensure reliable detection performance under critical conditions.
- **Skills:** Research Methodology, Python, EEG Signal Processing, Machine Learning.

Summer Intern, Neuroergonomics Lab, ISAE SUPAERO, Toulouse **Jul 2024**

- Worked with 8-electrode EEG data collection and analysis to verify a prediction algorithm.
- Supported team in EEG setup and data collection for studying spatial attention in simulated flight conditions.
- **Skills:** Python, Log Analysis.

Software/Firmware Engineer, Hewlett Packard, Bangalore **Oct 2020 – Aug 2023**

- Worked as Front-end engineer for Control Panel UI development in enterprise printers (40+ products) while focusing on reliable user interaction and system-level behavior.
- Acted as point-of-contact for feature rollouts and product line deployments, performing system triage, debugging, and customer escalation handling.
- Contributed to requirement analysis and cross-functional coordination to translate user and system needs into implementable specifications.
- **Skills:** JavaScript, C#, UI Development, System Debugging, Requirement Analysis.

Projects

Research Project : Low Cognitive Load Interface Design (Drone Swarm Teleoperation), ISAE-SUPAERO **Dec 2025-Feb 2026**

- Conducting a structured literature survey on interaction techniques for drone and robot swarm teleoperation, with a focused review of how cognitive load is assessed and reported.
- Deriving design insights and interface recommendations for low-cognitive-load swarm teleoperation, including key information to share and effective interaction modalities.
- **Skills:** Literature review, LaTeX, Zotero.

Autonomous Drone, ISAE-SUPAERO **Nov 2024 - Feb 2025**

- Designed a skill-based architecture with AI planning in the ROS 2 framework.
- Implemented autonomous navigation, object detection, and mission execution.
- **Skills:** ROS 2, AI Planning, Gazebo, Autonomous Systems.

Education

M.Sc in Aerospace Engineering (Embedded Systems), ISAE-SUPAERO, France **2023-2026**

- Relevant Courses: Instrumentation and Flight Data Analysis, Information Engineering, AI and Autonomous System, Real-Time Operating System.

B.E in Electronics and Communication Engineering, The National Institute of Engineering, Mysuru, India **2016-2020**

- Relevant Courses: Digital Electronics, Signals and Systems, Digital Signal Processing, Embedded Systems.

Scientific Engagement

- **ICCAS 2026, International Conference on Aeronautical Cognitive Systems (ISAE-SUPAERO, Toulouse)**: Volunteer and Participant.
- **BR41N.IO Brain-Computer Interface Designer's Hackathon 2026**: Participated in a 24-hour neurotechnology hackathon focused on stroke rehabilitation data analysis, improving EEG signal preprocessing using Common Average Reference (CAR) and CSP-based feature ranking.
- **CIME 2025, Interdisciplinary Congress on Extreme Worlds (ISAE-SUPAERO, Toulouse)**: Volunteer and Participant at an interdisciplinary congress focused on cognition, human performance, and adaptation in extreme environments.
- **Embedded Systems Cohort, ISAE-SUPAERO (2024–2025)**: Student Representative.
- **IMAV 2024, International Micro Air Vehicle Conference (Bristol, United Kingdom)**: Presented a colleague's work "On the Lateral Non-Stabilizability of PID Controllers for Vision-Based Line-Following Multicopters through Roll Angle Setpoints"
- **ICCAS 2024, International Conference on Aeronautical Cognitive Systems (ISAE-SUPAERO, Toulouse)**: Attendee.
- **UCSP (Ubiquitous Computing and Signal Processing), 2019–2020 (NIE, Mysuru)**: Team Leader of a student research initiative promoting undergraduate innovation through technical projects, scientific writing, workshops, and electronics and communication conferences.
- **ECE Association, 2017 (NIE, Mysuru)**: Student Representative.
- **Boeing National Aeromodelling Competition 2017 (IIT Kharagpur)**: Participant in a national aeromodelling competition; designed and built a fixed-wing aircraft model with gliding and object-dropping mechanisms. "'

Skills and Interests

- **Technical Expertise:** EEG Signal Processing, Computer Vision, Machine Learning, Research Methodology
- **Programming:** Python, C, C++, JavaScript
- **Tools:** MATLAB, Visual Studio, CMake, GitHub, JIRA, Zotero, LaTeX
- **Operating Systems:** Linux, Windows, RTOS
- **Soft Skills:** Critical Thinking, Curiosity, Autonomy, Problem Solving, Collaboration
- **Languages:** English (Fluent, IELTS 8.0), French (A2), Hindi (Professional Proficiency), Kannada (Native)
- **Interests:** Badminton, Swimming, Travel, Cooking

Publications

- **On the Lateral Non-Stabilizability of PID Controllers for Vision-Based Line-Following Multicopters through Roll Angle Setpoints.** P. Van Holm, A. B. Mallikarjuna, C. Simon Soria, and L. R. Lustosa. In *Proceedings of the 15th Annual International Micro Air Vehicle Conference and Competition (IMAV 2024)*, Bristol, United Kingdom, Sept. 16-20, 2024, pp. 277–284. Paper no. IMAV2024-33.
- **Pseudonymization of Text and Image Data to Provide Confidentiality.** Manthan, C. S., Roopa, K., Bindu, H. S., Apoorva, B. M., Madar, Mala D. *Proceedings of International Conference on Computational Intelligence, Data Science and Cloud Computing, Springer Singapore, 2021.* pp. 553–564. DOI: https://doi.org/10.1007/978-981-33-4968-1_43

References

- **Dr. Vsevolod Peysakhovich** , Associate Professor, Centre of Neuroergonomics and Human Factors, DCAS, ISAE SUPAERO.
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- **Dr. Michael Lauer** , Head of the Embedded Systems Major, Master of Aerospace Engineering, ISAE SUPAERO.
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